REWARD-TOKEN EXTREMIZATION IN BEST-OF-N TRAJECTORY TRANSFORMER PLANNING

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ABSTRACT

Trajectory Transformers plan by decoding interleaved state, action, and reward tokens. This creates a specific risk for Best-of-N or beam-style inference: as candidate count grows, selection pressure can improve the predicted reward-token sum while moving the selected token string into low-support, dynamically inconsistent regions. We study this mechanism in a reproducible discretized autoregressive surrogate. The main diagnostic treats the whole trajectory token string as the object under selection and measures trajectory log-likelihood, maximum prefix surprise, and token-level dynamics inconsistency. We give an exact finite-N law for score-selected candidate utility, show a controlled reward-token extremization failure, and evaluate a Support-Calibrated Plan Sieve that filters or downweights unsafe high-N candidates. The evidence is synthetic and mechanism-focused; benchmark neural Trajectory Transformer validation is left as the next step.

1 INTRODUCTION

Sequence modeling has become a compelling lens for offline reinforcement learning. Decision Transformers condition action generation on desired return (Chen et al., 2021), while Trajectory Transformers (TTs) model full trajectories and decode high-reward continuations with beam-style planning (Janner et al., 2021). This paper studies a narrow but important failure mode: what happens when the search budget is increased at inference time?

Best-of-N selection is attractive because it is simple: sample or decode many candidates, score them, and execute the best. In preference learning, related procedures can overoptimize imperfect reward models (Gao et al., 2023; Jinnai et al., 2024). For TTs, the problem is more structured. The reward being optimized is not only an external scalar; reward tokens are part of the autoregressive sequence alongside state and action tokens. This suggests a TT-specific failure: increasing N can select a string whose reward tokens are extreme even though its prefixes are surprising and its decoded state transitions do not match the dynamics implied by its decoded actions.

We call this *reward-token extremization*. The contribution is a diagnostic and repair for this architecture-specific mechanism, not a claim about broad benchmark performance.

2 SETUP

A trajectory token string has the repeated layout

$$(s_0, a_0, r_0, s_1, a_1, r_1, \ldots, s_{H-1}, a_{H-1}, r_{H-1}).$$

The surrogate model factorizes it autoregressively,

$$p_\theta(x_{1:3H}) = \prod_{t=1}^{3H} p_\theta(x_t \mid x_{<t}),$$

using smoothed context counts over discretized state, action, and reward vocabularies. This n-gram surrogate is intentionally inspectable. It preserves the TT planning interface while keeping every diagnostic reproducible.

Candidate score is the decoded reward-token sum

$$S(x) = \sum_{t=0}^{H-1} \hat{r}_t.$$

Realized utility $U(x)$ is measured by executing the decoded action sequence in the ground-truth simulator from the same initial state. The diagnostic also records average token log-likelihood, maximum prefix negative log-probability, and a dynamics inconsistency score:

$$D(x) = \frac{1}{H-1} \sum_{t=0}^{H-2} \left| \hat{s}_{t+1} - f(\hat{s}_t, \hat{a}_t) \right|.$$

3 Finite-N Selection Law

Theorem 1 (Exact score-selected utility law). *Let candidates $X_1, \ldots, X_N$ be i.i.d. from a finite distribution over outcomes i with probability p_i, proxy score s_i, and realized utility u_i. Select a maximum-score candidate and break ties uniformly among tied samples. Define $p_{=i} = \Pr(s(X) = s_i)$ and $p_{<i} = \Pr(s(X) < s_i)$. Then*

$$\Pr(\hat{X}_N = i) = N p_i \sum_{m=0}^{N-1} \binom{N-1}{m} \frac{p_{=i}^m p_{<i}^{N-1-m}}{m+1}.$$

Consequently,

$$\mathbb{E}[U(\hat{X}_N)] = \sum_i u_i \Pr(\hat{X}_N = i).$$

The proof distinguishes one sampled copy of outcome i. It is selected when no other sample has a higher score; if m other samples tie its score, the distinguished copy wins tie-breaking with probability $1/(m+1)$. The remaining samples must have equal or lower score.

Corollary 1 (Tail misalignment can reverse utility). *If increasing N increases the probability of selecting a high-score tail whose conditional realized utility is lower than the lower-score mass, then $\mathbb{E}[U(\hat{X}_N)]$ can decrease with N.*

The corollary is conditional. It does not say more search is always harmful. It says that proxy-score tails need utility alignment, not only high proxy score.

4 Support-Calibrated Plan Sieve

The repair calibrates thresholds on offline training trajectories. A candidate is accepted only if its average log-likelihood is above a low quantile, its maximum prefix surprise is below a high quantile, and its dynamics inconsistency is below a high quantile. Accepted candidates are ranked by a calibrated score:

$$S_{\text{sieve}}(x) = S(x) - \lambda_\ell[\tau_\ell - \ell(x)]_+ - \lambda_p[P(x) - \tau_p]_+ - \lambda_d[D(x) - \tau_d]_+.$$

If no candidate is accepted, the planner returns the lowest-risk candidate and marks the candidate set as blocked. An adaptive-N gate stops increasing N once the candidate risk rate exceeds the calibrated ceiling.

The oracle real-utility selector appears only as an upper-bound diagnostic and is not a deployable method.

5 Experiments

The environment is a one-dimensional control problem with a support-limited high-return corridor. Offline behavior mostly visits safe moderate-return actions and rarely visits high-return trajectories. The model is trained only from this offline data. We run $N \in \{1, 2, 4, 8, 16, 32, 64\}$ and evaluate four controls: in-support, out-of-support, anti-aligned scorer, and horizon stress.

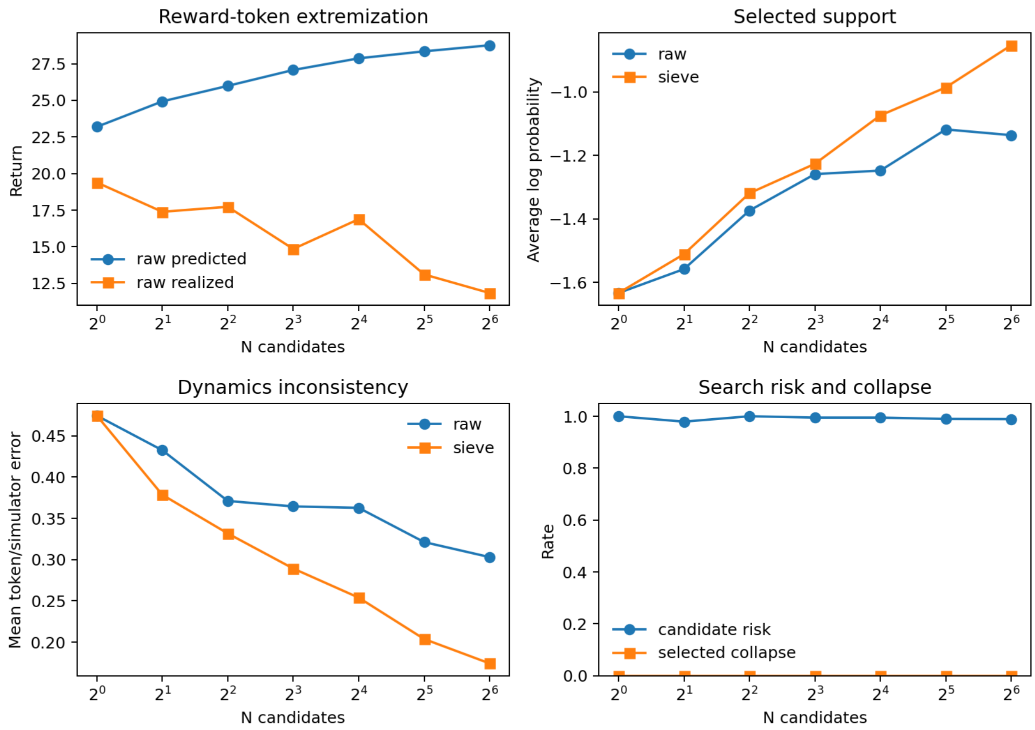

Figure 1: Horizon-stress control. Raw Best-of-N raises predicted reward-token return while selected support and dynamics consistency deteriorate. The sieve keeps selected plans closer to calibrated support.

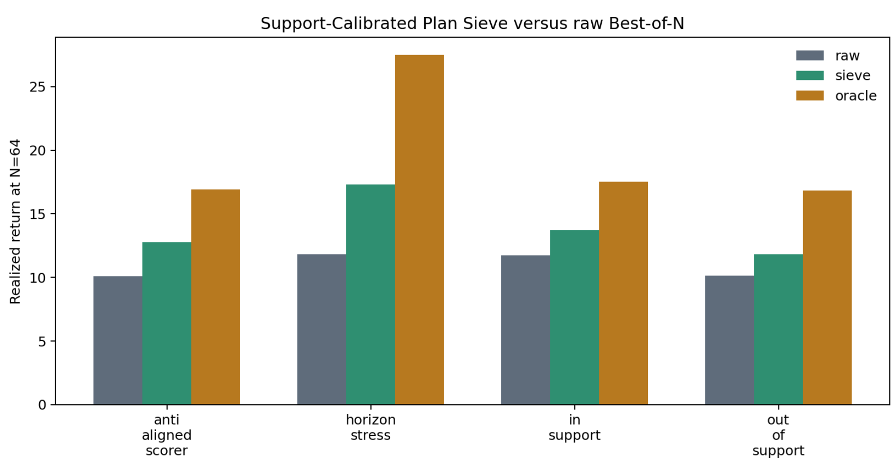

Figure 2: Realized return at $N = 64$. The oracle is an upper-bound diagnostic; the deployable comparison is raw Best-of-N versus the Support-Calibrated Plan Sieve.

6 Discussion

The results support a mechanism claim: Best-of-N TT planning can optimize reward tokens into low-support token strings. The key observation is that the selected object is not merely an action plan but an autoregressive sequence whose prefixes and state-token transitions can be audited. A generic reward reranker would miss that structure.

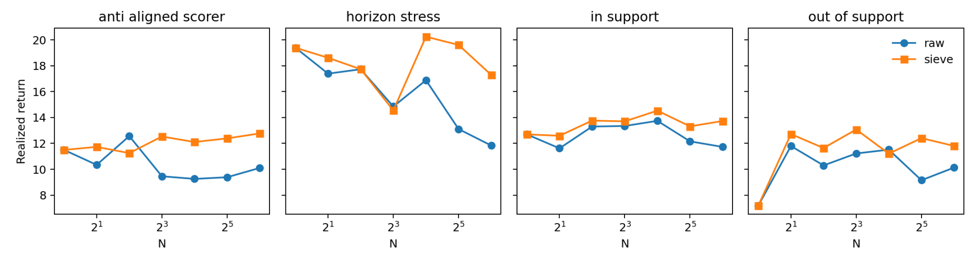

Figure 3: Controls across N. The in-support setting is less fragile, while out-of-support and horizon-stress settings expose the selection-risk mechanism.

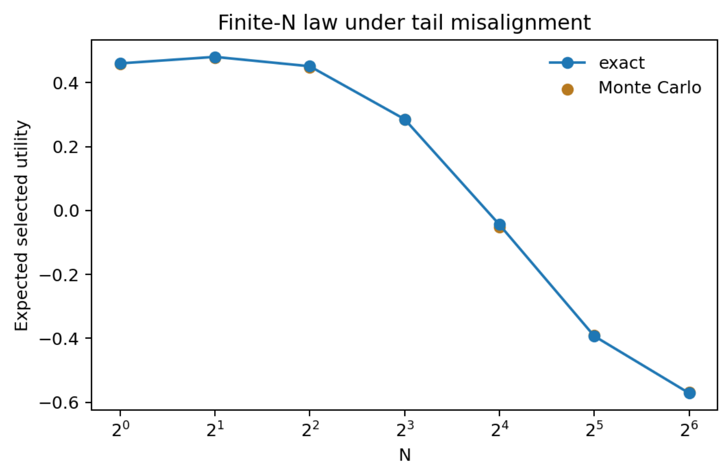

Figure 4: The finite-N law matches Monte Carlo in a tail-misalignment example where larger N lowers expected selected utility.

The main limitation is scale. The current implementation is synthetic and uses an interpretable surrogate rather than a neural Transformer. This is appropriate for an adversarial first pass because exact probabilities and support diagnostics can be inspected, but it is not a substitute for neural TT experiments on standard offline RL benchmarks. Future work should train a TT on D4RL-style data, compare sampling and deterministic beam variants, and test whether the same prefix-surprise and dynamics-consistency signals predict failure in larger models.

7 Reproducibility

The repository includes the simulator, token model, experiments, figures, tests, claim audit, proof audit, and this paper source. The primary commands are:

```
python -m trajectory_transformer_best_of_n.experiments.run_smoke
python -m trajectory_transformer_best_of_n.experiments.run_all
pytest
```

The claim-audit entry point is `run_claim_audit` in the same experiments package.

References

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A Claim Boundary

The paper makes a synthetic mechanism claim. It does not claim top benchmark performance, deployment readiness, or universal failure for every Trajectory Transformer. The final audit in the repository lists the strongest result, the strongest repair result, and the remaining weaknesses.